Bangladesh University of Business and Technology (BUBT)
Department of Computer Science and Engineering

Lightweight Hybrid CNN–Transformer for Imbalanced Diabetic Retinopathy Grading on RetinaMNIST

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Presented at BUBT CV Competition Poster Session 2026

Abstract

A lightweight hybrid CNN–Transformer for diabetic retinopathy grading on RetinaMNIST: a CNN captures local lesions, a linear-attention Transformer captures global context. It beats MedMNIST v2 baselines with 58.0% accuracy and 0.832 AUC.

Introduction

- DR is a major cause of vision loss; grading enables **early screening**.
- **Class imbalance**: 486 grade-0 vs 66 grade-4 images.
- Adjacent grades look alike, so **local lesions** and **global context** both matter.

Objectives

- Develop a lightweight hybrid CNN–Transformer for DR grading.
- Handle imbalance with class-aware augmentation and ordinal focal loss.
- Stabilise training with EMA, SWA and head tuning.

Literature Review

- **MedMNIST v2** (Yang et al., 2023): lightweight benchmark; baselines reach ≤ 53.1% accuracy.
- **SE-Net** (Hu et al., 2018): channel attention for stronger local features.
- **ViT** (Dosovitskiy et al., 2021): global context via patch self-attention.
- **Linear attention** (Katharopoulos et al., 2020): O(N) cost.
- **Class-balanced loss** (Cui et al., 2019) and **decoupled head tuning** (Kang et al., 2020) handle imbalance.

Methodology

1. Dataset
2. Preprocessing
3. Augmentation
4. Model Training
5. Validation
6. Testing



Dataset (fundus images): illustrative samples, grades 0–4

Model Architecture



Experimental Setup

Parameter	Value
Input size	192 × 192 (CLAHE)
Optimizer	AdamW (wd 1e-3)
Learning rate	2e-4, warm-up 5 + cosine
Epochs	150 max, stopped at 114
Loss	Ordinal focal (γ = 1.5)
Regularisation	EMA 0.99 (+ SWA tried)
Training	Scratch + head tuning

Results

Test metrics

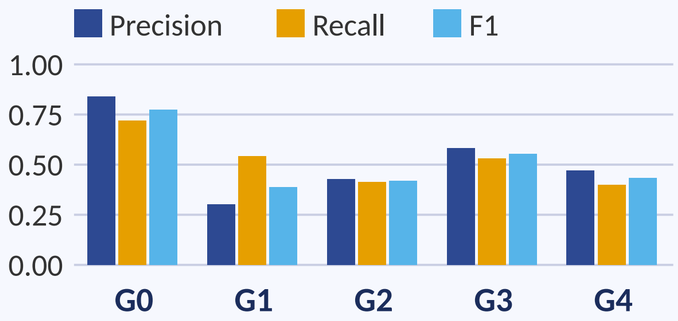
Metric	Value
Accuracy	**0.580**
Precision (macro)	**0.524**
Recall (macro)	**0.521**
F1-score (macro)	**0.514**
Balanced Acc.	**0.521**
QWK	**0.728**
AUC (OvR)	**0.832**

Confusion matrix (true × predicted)

	P0	P1	P2	P3	P4
T0	125	31	15	3	0
T1	9	25	9	2	1
T2	12	24	38	14	4
T3	2	3	23	36	4
T4	1	0	4	7	8

Darker = higher share of true grade. Most errors are adjacent.

Per-class test performance (0–1)



Comparison with MedMNIST v2

Method	AUC	ACC
ResNet-18 (28)	**0.717**	**0.524**
ResNet-18 (224)	**0.710**	**0.493**
ResNet-50 (28)	**0.726**	**0.528**
Google AutoML Vision	**0.750**	**0.531**
Ours (7.99 M)	**0.832**	**0.580**

Baselines: MedMNIST v2 (28 / 224 px). Ours: 192 px, from scratch.

Discussion

Findings

- Grade 0 is best (F1 0.77); grade 1 is hardest.
- QWK 0.728, AUC 0.832: 72% of errors are adjacent grades.
- Class-aware training lifts grade-1 recall to 0.54.

Limitations

- Small dataset, single seed, no k-fold CV.
- Grades 1–2 confused (grade-1 precision 0.30).

Conclusion & Future Work

- **Lightweight (7.99 M):** SE-CNN for local lesions, linear attention for global context.
- **Imbalance- and ordinal-aware training** keeps most errors adjacent.
- **Future:** multi-seed ensembles, k-fold CV, self-supervised pre-training.

Thanks & Contact

Thanks to the BUBT CV Competition organisers and the MedMNIST team.

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Scan for paper & code

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